LLM Cost Estimate for Literary Analysis Project

Per-Poem Token Estimates

Input: Average Poem

- **Average word count:** 172 words
- **Average characters per word:** 5
- **Total characters:** 172 × 6 = 1,032 characters
- **Tokens (1 token ≈ 4 chars):** ~258 tokens

Prompt 1: Literary Devices Inference

- Instruction text: ~50 tokens
- Literary devices list (35 devices): ~150 tokens
- Poem text: ~258 tokens
- **Total input:** ~458 tokens per poem

Prompt 2: Score Inference

- Instruction & rubric (5 categories with descriptions): ~150 tokens
- Poem text: ~258 tokens
- **Total input:** ~408 tokens per poem

Expected Outputs (per inference)

- **Literary Devices:** Rationale (~200 tokens) + device list (~20 tokens) = ~220 tokens
- **Score:** Rationale (~300 tokens) + 5 numerical scores (~10 tokens) = ~310 tokens

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Cost Breakdown per Poem (2 inferences × 3 models = 6 API calls)

Input Tokens (all models)

- Literary Devices: 458 tokens × 3 models = 1,374 tokens
- Scoring: 408 tokens × 3 models = 1,224 tokens
- **Total input: 2,598 tokens per poem**

Output Tokens (estimates)

- Literary Devices: 220 tokens × 3 models = 660 tokens
- Scoring: 310 tokens × 3 models = 930 tokens
- **Total output: 1,590 tokens per poem**

Pricing (estimated 2026 rates)

Model Pricing (per 1M tokens)

- **GPT-5.4:** Input $2.50, Output $15.00
- **Claude 4.6 Opus:** Input $5.00, Output $25.00
- **Gemini 3.1 Pro:** Input $2.00, Output $12.00

Cost per Poem (per model)

Model	Input Cost ($ per token)	Output Cost ($ per token)	Total Cost
GPT-5.4	$0.0000025	$0.000015	**$0.030345**
Claude Opus	$0.000005	$0.000025	**$0.05274**
Gemini Pro	$0.000002	$0.000012	**$0.024276**

Cost per Poem (all 3 models): ~$0.107361

Scaling to Full Dataset

Poem Count	Total Cost	GPT-5.4	Claude	Gemini	Notes
10	$1.07361	$0.30345	$0.52740	$0.24276	Test batch
50	$5.36805	$1.51725	$2.63700	$1.21380	
100	$10.73610	$3.03450	$5.27400	$2.42760	
500	$53.68050	$15.17250	$26.37000	$12.13800	
1,000	$107.36100	$30.34500	$52.74000	$24.27600	Full-scale run

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Full-Scale Allocation With Leeway (1,000 Poems)

Baseline full-scale API cost is **$107.36**.

Budget by Leeway Level

Leeway	Formula	Total Budget
0%	$107.36 × 1.00	**$107.36**
15%	$107.36 × 1.15	**$123.47**
25%	$107.36 × 1.25	**$134.20**
40%	$107.36 × 1.40	**$150.31**

Specific Allocation (Recommended: 25% Leeway)

Using a **$134.20** budget cap:

- **Core planned inference (all poems, all models):** $107.36
- **Retry / transient API failures reserve (10%):** $13.42

- **Longer-than-expected outputs reserve (7.5%):** $8.05
- **Prompt iteration / QA spot-check reserve (5%):** $5.37

Total allocated: **$134.20**

Model-Level Allocation at 25% Leeway

Model	Baseline	With 25% Leeway
GPT-5.4	$30.35	$37.93
Claude 4.6 Opus	$52.74	$65.93
Gemini 3.1 Pro	$24.28	$30.35
Total	**$107.36**	**$134.20**

Note: Current `data/poems.csv` has 1 poem row, so this is a projection for the full target dataset.